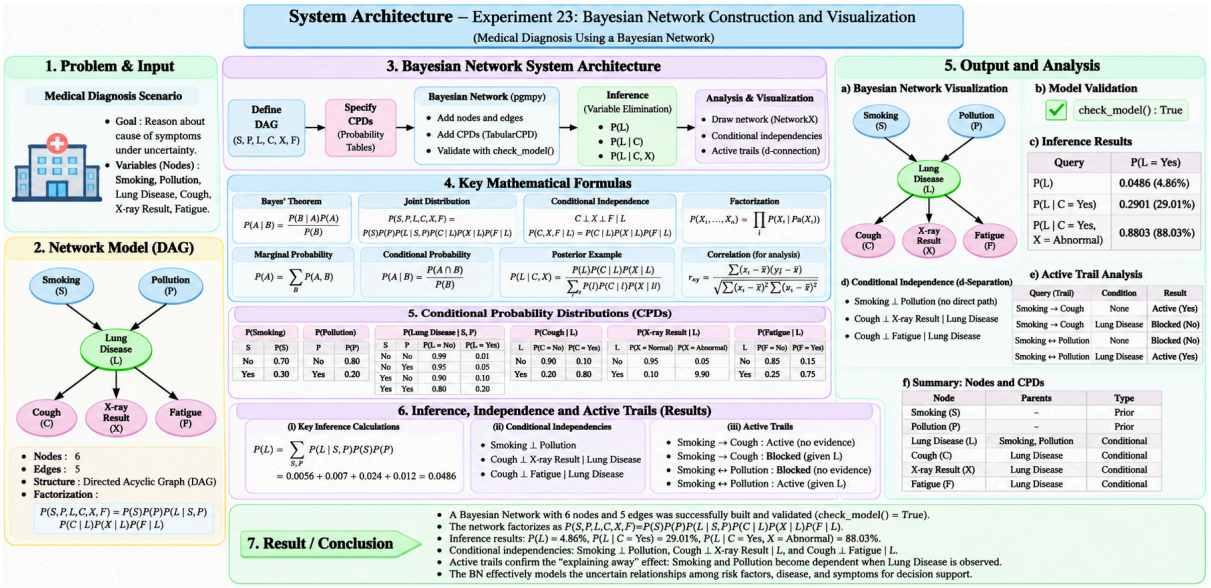
Experiment 23: Bayesian Network Construction and Visualization

1. Aim

To construct and visualize a Bayesian Network for medical diagnosis using the pgmpy library. To define a Directed Acyclic Graph (DAG) containing Smoking, Pollution, Lung Disease, Cough, X-ray Result, and Fatigue, specify Conditional Probability Distributions (CPDs) for every node, validate the network using `check_model()`, and identify conditional independencies and active trails. The experiment uses pgmpy, NetworkX, and Matplotlib to understand probabilistic relationships between medical conditions and symptoms.



2. Mathematical Formulas

1. Bayes' Theorem

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

2. Joint Probability Distribution

$$\boxed{P(S, P, L, C, X, F) = P(S)P(P)P(L|S, P)P(C|L)P(X|L)P(F|L)}$$

Where:

- S = Smoking
- P = Pollution
- L = Lung Disease
- C = Cough
- X = X ray Result
- F = Fatigue

3. Conditional Probability

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

4. Marginal Probability

$$\boxed{P(A) = \sum_{B} P(A, B)}$$

5. Conditional Probability Distribution (CPD)

$$\boxed{P(X|Parents(X))}$$

Each node has a probability distribution based on its parent nodes.

6. Probability of Lung Disease

$$\boxed{P(L|S, P)}$$

7. Conditional Independence

$$\boxed{P(C, X, F|L) = P(C|L)P(X|L)P(F|L)}$$

This represents conditional independence of symptoms given lung disease.

8. Network Factorization

$$\boxed{P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i|Parents(X_i))}$$

9. Posterior Probability

$$\boxed{P(L|C, X, F) = \frac{P(L, C, X, F)}{P(C, X, F)}}$$

10. Conditional Independence Notation

$$\boxed{A \perp B \mid C}$$

This means A and B are conditionally independent given C.

CODE

```
# Experiment 23: Bayesian Network Construction and Visualization
!pip install -q pgmpy networkx
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import networkx as nx
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination
# 1. Define the Bayesian Network Structure
model = DiscreteBayesianNetwork([
    ("Smoking","LungDisease"),
    ("Pollution","LungDisease"),
    ("LungDisease","Cough"),
    ("LungDisease","XrayResult"),
    ("LungDisease","Fatigue")
])
# 2. Define Conditional Probability Distributions
cpd_smoking = TabularCPD(
    variable="Smoking",
    variable_card=2,
    values=[[0.7],[0.3]],
    state_names={"Smoking":["No","Yes"]}
)
cpd_pollution = TabularCPD(
    variable="Pollution",
    variable_card=2,
    values=[[0.8],[0.2]],
    state_names={"Pollution":["No","Yes"]}
)
cpd_lung = TabularCPD(
    variable="LungDisease",
    variable_card=2,
    values=[
        [0.99,0.95,0.90,0.80],
        [0.01,0.05,0.10,0.20]
    ],
    evidence=["Smoking","Pollution"],
    evidence_card=[2,2],
    state_names={
        "LungDisease":["No","Yes"],
        "Smoking":["No","Yes"],
        "Pollution":["No","Yes"]
    }
)
cpd_cough = TabularCPD(
    variable="Cough",
    variable_card=2,
    values=[[0.9,0.2],[0.1,0.8]],
    evidence=["LungDisease"],
    evidence_card=[2],
    state_names={
        "Cough":["No","Yes"],
        "LungDisease":["No","Yes"]
    }
)
cpd_xray = TabularCPD(
```

```
    variable="XrayResult",
    variable_card=2,
    values=[[0.95,0.10],[0.05,0.90]],
    evidence=["LungDisease"],
    evidence_card=[2],
    state_names={
        "XrayResult":["Normal","Abnormal"],
        "LungDisease":["No","Yes"]
    }
)
cpd_fatigue = TabularCPD(
    variable="Fatigue",
    variable_card=2,
    values=[[0.85,0.25],[0.15,0.75]],
    evidence=["LungDisease"],
    evidence_card=[2],
    state_names={
        "Fatigue":["No","Yes"],
        "LungDisease":["No","Yes"]
    }
)
# 3. Add All CPDs to the Network
model.add_cpds(
    cpd_smoking,
    cpd_pollution,
    cpd_lung,
    cpd_cough,
    cpd_xray,
    cpd_fatigue
)
# 4. Validate the Bayesian Network
print("MODEL VALIDATION")
print("Is the model valid?",model.check_model())
print("\nBayesian Network Nodes:")
print(list(model.nodes()))
print("\nBayesian Network Edges:")
print(list(model.edges()))
# 5. Display Conditional Probability Tables
print("\nCONDITIONAL PROBABILITY DISTRIBUTIONS")
for cpd in model.get_cpds():
    print("\n",cpd)
# 6. Visualize the Bayesian Network
G = nx.DiGraph()
G.add_nodes_from(model.nodes())
G.add_edges_from(model.edges())
pos = {
    "Smoking":(0,2),
    "Pollution":(2,2),
    "LungDisease":(1,1),
    "Cough":(0,0),
    "XrayResult":(1,0),
    "Fatigue":(2,0)
}
plt.figure(figsize=(10,7))
nx.draw(
    G,pos,
    with_labels=True,
    node_size=3500,
```

```
    node_color="lightblue",
    font_size=10,
    font_weight="bold",
    arrows=True,
    arrowsize=25,
    edge_color="gray"
)
plt.title("Bayesian Network for Medical Diagnosis")
plt.show()
# 7. Perform Probabilistic Inference
inference = VariableElimination(model)
# Probability of Lung Disease
result1 = inference.query(
    variables=["LungDisease"]
)
print("\nPROBABILITY OF LUNG DISEASE")
print(result1)
# Probability of Lung Disease given Cough
result2 = inference.query(
    variables=["LungDisease"],
    evidence={"Cough":"Yes"}
)
print("\nLUNG DISEASE GIVEN COUGH")
print(result2)
# Probability of Lung Disease given Cough and Abnormal X-ray
result3 = inference.query(
    variables=["LungDisease"],
    evidence={
        "Cough":"Yes",
        "XrayResult":"Abnormal"
    }
)
print("\nLUNG DISEASE GIVEN COUGH AND ABNORMAL X-RAY")
print(result3)
# 8. Conditional Independence Analysis
print("\nCONDITIONAL INDEPENDENCE")
print("Smoking independent of Pollution:",
      not model.is_dconnected("Smoking","Pollution"))
print("Cough independent of X-ray given Lung Disease:",
      not model.is_dconnected(
          "Cough","XrayResult",
          observed=["LungDisease"]
      ))
print("Cough independent of Fatigue given Lung Disease:",
      not model.is_dconnected(
          "Cough","Fatigue",
          observed=["LungDisease"]
      ))
# 9. Active Trail Analysis
print("\nACTIVE TRAIL ANALYSIS")
print("Smoking to Cough without evidence:",
      model.is_dconnected("Smoking","Cough"))
print("Smoking to Cough given Lung Disease:",
      model.is_dconnected(
          "Smoking","Cough",
          observed=["LungDisease"]
      ))
print("Smoking to Pollution given Lung Disease:",
```

```
      model.is_dconnected(
          "Smoking","Pollution",
          observed=["LungDisease"]
      ))
# 10. Display Conditional Probability Summary
summary = pd.DataFrame({
    "Node":[
        "Smoking","Pollution","Lung Disease",
        "Cough","X-ray Result","Fatigue"
    ],
    "Parent Nodes":[
        "None","None","Smoking, Pollution",
        "Lung Disease","Lung Disease","Lung Disease"
    ],
    "Type":[
        "Prior","Prior","Conditional",
        "Conditional","Conditional","Conditional"
    ]
})
print("\nNETWORK SUMMARY")
print(summary.to_string(index=False))
print("\nEXPERIMENT 23 COMPLETED SUCCESSFULLY")
```

OUTPUT

```
Bayesian Network Nodes:
['Smoking', 'LungDisease', 'Pollution', 'Cough', 'XrayResult', 'Fatigue']

Bayesian Network Edges:
[('Smoking', 'LungDisease'), ('LungDisease', 'Cough'), ('LungDisease', 'XrayResult'), ('LungDisease', 'Fatigue'), ('Pollution', 'LungDisease')]

CONDITIONAL PROBABILITY DISTRIBUTIONS

 +-------------+-----+
| Smoking(No)  | 0.7 |
+-------------+-----+
| Smoking(Yes) | 0.3 |
+-------------+-----+

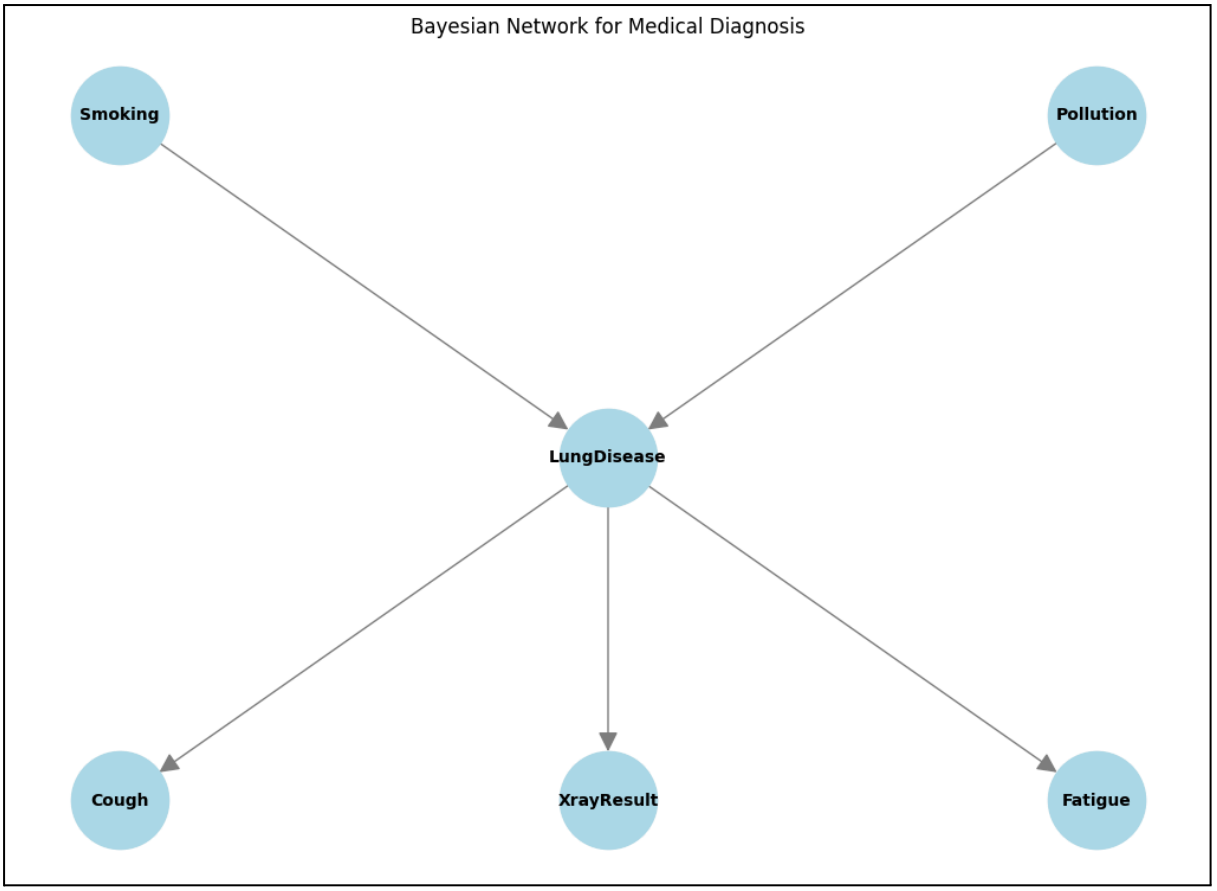
 +---------------+-----+
| Pollution(No)  | 0.8 |
+---------------+-----+
| Pollution(Yes) | 0.2 |
+---------------+-----+

 +-----------------+--------------+-----+--------------+---------------+
| Smoking          | Smoking(No)  | ... | Smoking(Yes) | Smoking(Yes)  |
+-----------------+--------------+-----+--------------+---------------+
| Pollution        | Pollution(No) | ... | Pollution(No) | Pollution(Yes) |
+-----------------+--------------+-----+--------------+---------------+
| LungDisease(No)  | 0.99         | ... | 0.9          | 0.8           |
+-----------------+--------------+-----+--------------+---------------+
| LungDisease(Yes) | 0.01         | ... | 0.1          | 0.2           |
+-----------------+--------------+-----+--------------+---------------+

 +------------+----------------+-----------------+
| LungDisease | LungDisease(No) | LungDisease(Yes) |
+------------+----------------+-----------------+
| Cough(No)   | 0.9            | 0.2             |
+------------+----------------+-----------------+
| Cough(Yes)  | 0.1            | 0.8             |
+------------+----------------+-----------------+

 +---------------------+----------------+-----------------+
| LungDisease          | LungDisease(No) | LungDisease(Yes) |
+---------------------+----------------+-----------------+
| XrayResult(Normal)   | 0.95           | 0.1             |
+---------------------+----------------+-----------------+
| XrayResult(Abnormal) | 0.05           | 0.9             |
+---------------------+----------------+-----------------+

 +-------------+----------------+-----------------+
| LungDisease  | LungDisease(No) | LungDisease(Yes) |
+-------------+----------------+-----------------+
| Fatigue(No)  | 0.85           | 0.25            |
+-------------+----------------+-----------------+
| Fatigue(Yes) | 0.15           | 0.75            |
+-------------+----------------+-----------------+
```


Bayesian Network for Medical Diagnosis
Smoking
Pollution
LungDisease
Cough
XrayResult
Fatigue

```
PROBABILITY OF LUNG DISEASE
+------------------+--------------------+
| LungDisease      |   phi(LungDisease) |
+==================+====================+
| LungDisease(No)  |             0.9514 |
+------------------+--------------------+
| LungDisease(Yes) |             0.0486 |
+------------------+--------------------+

LUNG DISEASE GIVEN COUGH
+------------------+--------------------+
| LungDisease      |   phi(LungDisease) |
+==================+====================+
| LungDisease(No)  |             0.7099 |
+------------------+--------------------+
| LungDisease(Yes) |             0.2901 |
+------------------+--------------------+

LUNG DISEASE GIVEN COUGH AND ABNORMAL X-RAY
+------------------+--------------------+
| LungDisease      |   phi(LungDisease) |
+==================+====================+
| LungDisease(No)  |             0.1197 |
+------------------+--------------------+
| LungDisease(Yes) |             0.8803 |
+------------------+--------------------+

CONDITIONAL INDEPENDENCE
Smoking independent of Pollution: True
Cough independent of X-ray given Lung Disease: True
Cough independent of Fatigue given Lung Disease: True

ACTIVE TRAIL ANALYSIS
Smoking to Cough without evidence: True
Smoking to Cough given Lung Disease: False
Smoking to Pollution given Lung Disease: True

NETWORK SUMMARY
        Node       Parent Nodes         Type
     Smoking               None        Prior
   Pollution               None        Prior
Lung Disease Smoking, Pollution Conditional
       Cough       Lung Disease Conditional
X-ray Result       Lung Disease Conditional
     Fatigue       Lung Disease Conditional
```

5. RESULT

A Bayesian Network for medical diagnosis was successfully constructed using pgmpy. Six nodes representing Smoking, Pollution, Lung Disease, Cough, X-ray Result, and Fatigue were connected through a Directed Acyclic Graph. Conditional Probability Distributions were defined for every node, and the network was validated using `check_model()`. Probabilistic inference was performed to calculate the probability of lung disease under different symptom conditions. NetworkX was used to visualize the network, and conditional independencies and active trails were examined. Thus, the experiment demonstrates how Bayesian Networks represent uncertain relationships and support probabilistic decision-making in medical diagnosis.